158.337
GROUP PROJECT
GROUP 35
PART B
Queries (SQL/LINQ), Triggers, Procedures, Functions (PL/SQL) & MongoDB (NoSQL)

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Section A (SQL Queries)

Outputs are provided as both excel exports and snips, the latter to show that the output was not manufactured outside Oracle. Note outputs may vary from what is presented here if code is run in current database, as test records were added during section B.

Question Aa)

Write a query that lists both customer and author details (only their ids, first and last names). Provide suitable headings for the merged list:

Code:

```
SELECT
    TO_CHAR(Cust_Num) AS "ID" ,
    Cust_FName || ' ' || Cust_LName AS "Full Name"
    FROM CUSTOMER
UNION
SELECT
    Author_ID AS "ID",
    Author_FName || ' ' || Author_LName AS "Full Name"
    FROM AUTHOR
;
```

Output:

ID	Full Name
1001	BONITA MORALES
1002	RYAN THOMPSON
1003	LEILA SMITH
1004	THOMAS PIERSON
1005	CINDY GIRARD
1006	MESHIA CRUZ
1007	TAMMY GIANA
1008	KENNETH JONES
1009	JORGE PEREZ
1010	JAKE LUCAS
1011	REESE MCGOVERN
1012	WILLIAM MCKENZIE
1013	NICHOLAS NGUYEN
1014	JASMINE LEE
1015	STEVE SCHELL
1016	MICHELL DAUM
1017	BECCA NELSON
1018	GREG MONTIASA
1019	JENNIFER SMITH
1020	KENNETH FALAH
A100	JAMES AUSTIN

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A105	JUAN ADAMS
B100	JACK BAKER
D200	Aeneas Dardania
F100	OSCAR FIELDS
I200	Odysseus Ithica
J100	JANICE JONES
K100	TAMARA KZOCHSKY
M100	SHEILA MARTINEZ
M200	Agamemnon Mycenae
P100	LISA PORTER
P105	TINA PETERSON
R100	ROBERT ROBINSON
S100	SAM SMITH
W100	WILLIAM WHITE
W105	LISA WHITE
W110	ANTHONY WILKINSON

	ID	Full Name
1	1001	BONITA MORALES
2	1002	RYAN THOMPSON
3	1003	LEILA SMITH
4	1004	THOMAS PIERSON
5	1005	CINDY GIRARD
6	1006	MESHIA CRUZ
7	1007	TAMMY GIANA
8	1008	KENNETH JONES
9	1009	JORGE PEREZ
10	1010	JAKE LUCAS
11	1011	REESE MCGOVERN
12	1012	WILLIAM MCKENZIE
13	1013	NICHOLAS NGUYEN
14	1014	JASMINE LEE
15	1015	STEVE SCHELL
16	1016	MICHELL DAUM
17	1017	BECCA NELSON
18	1018	GREG MONTIASA
19	1019	JENNIFER SMITH
20	1020	KENNETH FALAH
21	A100	JAMES AUSTIN
22	A105	JUAN ADAMS
23	B100	JACK BAKER
24	D200	Aeneas Dardania
25	F100	OSCAR FIELDS
26	I200	Odysseus Ithica
27	J100	JANICE JONES
28	K100	TAMARA KZOCHSKY
29	M100	SHEILA MARTINEZ
30	M200	Agamemnon Mycenae
31	P100	LISA PORTER
32	P105	TINA PETERSON
33	R100	ROBERT ROBINSON
34	S100	SAM SMITH
35	W100	WILLIAM WHITE
36	W105	LISA WHITE
37	W110	ANTHONY WILKINSON

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Question Ab)

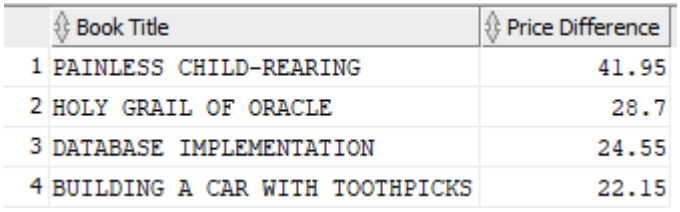
Write a query that will list the books with the price difference (between retail and cost) of $20 or more. Display your results in the decreasing order of the price difference

Code:

```
SELECT
    Book_Title AS "Book Title",
    Book_Retail - Book_Cost AS "Price Difference"
    FROM BOOK
WHERE
    Book_Retail - Book_Cost >= 20
ORDER BY "Price Difference" DESC
;
```

Output:

Book Title	Price Difference
PAINLESS CHILD-REARING	41.95
HOLY GRAIL OF ORACLE	28.7
DATABASE IMPLEMENTATION	24.55
BUILDING A CAR WITH TOOTHPICKS	22.15



	Book Title	Price Difference
1	PAINLESS CHILD-REARING	41.95
2	HOLY GRAIL OF ORACLE	28.7
3	DATABASE IMPLEMENTATION	24.55
4	BUILDING A CAR WITH TOOTHPICKS	22.15

Question Ac)

Write a query that will list books that have retail price of $35 or less and were published in any of the years 1999 or 2001. Display the year of publication values under the column titled as Publication Year. Show the query results in the increasing order of the publication year and decreasing retail price.

Code:

```
SELECT
    Book_Title AS "Book Title",
    Book_Retail AS "Retail Price",
    Book_Pubdate AS "Publication Year"
FROM BOOK
WHERE (
    Book_Retail <= 35
    AND (
        TO_CHAR(Book_Pubdate, 'YYYY') = '1999'
        OR TO_CHAR(Book_Pubdate, 'YYYY') = '2001'
        )
    )
ORDER BY
    "Publication Year" ASC,
    "Retail Price" DESC
;
```

Output:

Book Title	Retail Price	Publication Year
HOW TO MANAGE THE MANAGER	31.95	9/05/1999
BODYBUILD IN 10 MINUTES A DAY	30.95	21/01/2001
HANDCRANKED COMPUTERS	25	21/01/2001
BIG BEAR AND LITTLE DOVE	8.95	8/11/2001
REVENGE OF MICKEY	22	14/12/2001

	Book Title	Retail Price	Publication Year
1	HOW TO MANAGE THE MANAGER	31.95	09/05/99
2	BODYBUILD IN 10 MINUTES A DAY	30.95	21/01/01
3	HANDCRANKED COMPUTERS	25	21/01/01
4	BIG BEAR AND LITTLE DOVE	8.95	08/11/01
5	REVENGE OF MICKEY	22	14/12/01

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Question Ad)

Write a query that will list publishers (with details such as name) and total number of books published by them. Include only those publishers that have 2 or more books published and display your output in the decreasing order of total number of publications

Code:

```
SELECT
    PUBLISHER.Pub_Name AS "Publisher Name",
    COUNT(BOOK.Pub_ID) AS "Total Publications"
FROM BOOK
INNER JOIN
    Publisher
    ON
    PUBLISHER.Pub_ID = BOOK.Pub_ID
GROUP BY
    PUBLISHER.Pub_Name
    HAVING COUNT(BOOK.Pub_ID) >1
ORDER BY
    "Total Publications" DESC
;
```

Output:

Publisher Name	Total Publications
READING MATERIALS INC.	4
PRINTING IS US	4
REED-N-RITE	3
AMERICAN PUBLISHING	3
PUBLISH OUR WAY	2

	Publisher Name	Total Publications
1	READING MATERIALS INC.	4
2	PRINTING IS US	4
3	REED-N-RITE	3
4	AMERICAN PUBLISHING	3
5	PUBLISH OUR WAY	2

Question Ae)

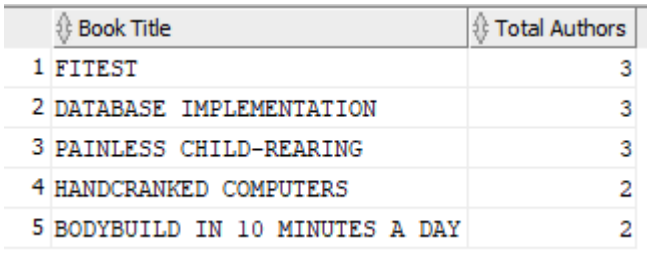
Write a query to list books that have multiple authors. Also, display the number of authors who wrote the book.

Code:

```
SELECT
    BOOK.Book_Title AS "Book Title",
    COUNT(BOOKAUTHOR.Book_ISBN) AS "Total Authors"
FROM BOOKAUTHOR
INNER JOIN
    BOOK
    ON
    BOOK.Book_ISBN = BOOKAUTHOR.Book_ISBN
GROUP BY
    BOOK.Book_Title
    HAVING COUNT(BOOKAUTHOR.Book_ISBN) >1
ORDER BY
    "Total Authors"
DESC
;
```

Output:

Book Title	Total Authors
DATABASE IMPLEMENTATION	3
PAINLESS CHILD-REARING	3
HANDCRANKED COMPUTERS	2
BODYBUILD IN 10 MINUTES A DAY	2



	Book Title	Total Authors
1	FITEST	3
2	DATABASE IMPLEMENTATION	3
3	PAINLESS CHILD-REARING	3
4	HANDCRANKED COMPUTERS	2
5	BODYBUILD IN 10 MINUTES A DAY	2

Question Af)

Write a query that will list books from the 'computer' category along with other details such as author(s), publisher, etc. List the book authors in a single row and the query should be able to accommodate all the casing-variations of the category values in the database.

Code:

Note, that because the question requirements were somewhat vague and only stipulated details such as authors, publisher, book title and category, there are shown in the code. The code provided is easily editable to include any other fields. Instructions to do so are shown as comments in the code.

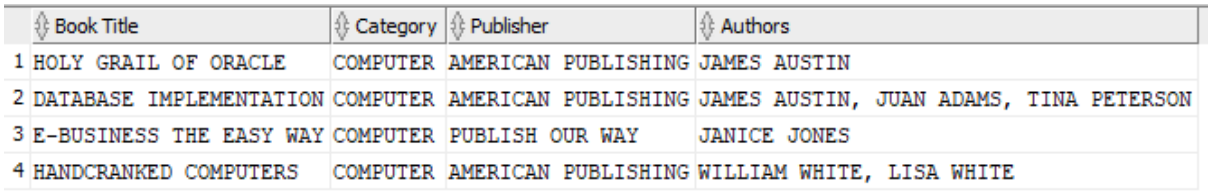
```
WITH A AS (
    SELECT
        BOOK.*, --allows any book attribute to be selected
        BOOKAUTHOR.*,
        AUTHOR.Author_Fname || ' ' || Author.Author_LName AS "Author",
        PUBLISHER.Pub_name
    FROM BOOK
    JOIN BOOKAUTHOR ON BOOK.Book_isbn = BOOKAUTHOR.book_isbn
    JOIN AUTHOR ON BOOKAUTHOR.Author_id = AUTHOR.author_id
    JOIN PUBLISHER ON BOOK.pub_id = PUBLISHER.pub_id
    WHERE Book_Category = 'COMPUTER'
    )

SELECT --specifically select and alias any book table value you want
    Book_title AS "Title",
    book_category AS "Category" ,
    pub_name AS "Publisher",
    LISTAGG("Author", ', ') AS "Authors"
FROM A
--include all selected book attributes in the group by statement to show them
in output
GROUP BY Book_title, book_category, pub_name
;
```


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Output:

Book Title	Category	Publisher	Authors
HOLY GRAIL OF ORACLE	COMPUTER	AMERICAN PUBLISHING	JAMES AUSTIN
DATABASE IMPLEMENTATION	COMPUTER	AMERICAN PUBLISHING	JAMES AUSTIN, JUAN ADAMS, TINA PETERSON
E-BUSINESS THE EASY WAY	COMPUTER	PUBLISH OUR WAY	JANICE JONES
HANDCRANKED COMPUTERS	COMPUTER	AMERICAN PUBLISHING	WILLIAM WHITE, LISA WHITE



	Book Title	Category	Publisher	Authors
1	HOLY GRAIL OF ORACLE	COMPUTER	AMERICAN PUBLISHING	JAMES AUSTIN
2	DATABASE IMPLEMENTATION	COMPUTER	AMERICAN PUBLISHING	JAMES AUSTIN, JUAN ADAMS, TINA PETERSON
3	E-BUSINESS THE EASY WAY	COMPUTER	PUBLISH OUR WAY	JANICE JONES
4	HANDCRANKED COMPUTERS	COMPUTER	AMERICAN PUBLISHING	WILLIAM WHITE, LISA WHITE

Question Ag)

Write a query that will list the customer who has ordered the greatest number of items (two copies of the same book will be counted as two items) to date. If there is more than one customer, then your query should list all those customers.

Code:

```
SELECT
    Customer.Cust_Num AS "Customer Number",
    Customer.Cust_FName || ' ' || Customer.Cust_LName AS "Customer Name",
    COUNT(BookOrderItem.BO_OrderNum) AS "Total Items Ordered"
FROM CUSTOMER
INNER JOIN
    BOOKORDER
    ON
    CUSTOMER.Cust_Num = BookOrder.Cust_Num
INNER JOIN
    BOOKORDERITEM
    ON
    BOOKORDER.BO_OrderNum = BOOKORDERITEM.BO_OrderNum
GROUP BY
    CUSTOMER.Cust_Num, CUSTOMER.Cust_FName, CUSTOMER.Cust_LName
    HAVING COUNT(BOOKORDERITEM.BO_OrderNum) = (
        SELECT MAX(ItemCount)
        FROM (SELECT
            COUNT(BOOKORDERITEM.BO_OrderNum) AS ItemCount
            FROM BOOKORDERITEM
GROUP BY BOOKORDERITEM.BO_OrderNum
        )
    )
;
```

Output:

Customer Number	Customer Name	Total Items Ordered
1017	BECCA NELSON	4

	Customer Number	Customer Name	Total Items Ordered
1	1017	BECCA NELSON	4

Question Ah)

Write a query that will display the customer that referred the greatest number of customers. Again, like g) there can be more than one customer.

Code:

```
SELECT
    C.Cust_referred_by AS "Customer Number",
    CUSTOMER.Cust_FName || ' ' || CUSTOMER.Cust_LName AS "Customer Name",
    COUNT(*) AS "Total Referrals"
FROM CUSTOMER
LEFT JOIN CUSTOMER C ON CUSTOMER.Cust_num = C.Cust_referred_by
GROUP BY C.Cust_referred_by, CUSTOMER.Cust_FName, CUSTOMER.Cust_LName
    HAVING COUNT(*) = (
        SELECT MAX(ReferralCount)
        FROM (
            SELECT
                COUNT(*) AS ReferralCount
            FROM CUSTOMER
            WHERE
                Cust_Referred_by IS NOT NULL
            GROUP BY
                Cust_Referred_by
        )
    )
;
```

Output:

Customer Number	Customer Name	Total Referrals
1003	LEILA SMITH	3

	Customer N...	Customer Name	Total Referrals
1	1003	LEILA SMITH	3

Section B (PL SQL)

Question Ba)

Think of some useful business rule or situation (based on the provided database) where it would be appropriate for your trigger to fire…

The business process targeted is sales, which has a direct link to inventory management.
The trigger created does f things when a new book order item is created.

1. When a sale is added as an order item the cost is calculated and added to the total order.
2. The number of items added are reduced from the book's stock level.
3. If the stock level goes below the minimum stock level, a flag to reorder is raised by setting the 'reorder' value to 1. A system message is printed.
4. If the order would put the stock below 0, then an error is thrown indicating there is not enough stock. This was decided as opposed to triggering a backorder process, which would be a separate business process requiring a customer decision point; to accept longer delivery times or not.
5. If the order total in the BOOKORDER table becomes over $200 a 10% discount is applied. The discount if applied only once for the first time this condition is met, else the discount could become very large. This is achieved, by requiring the 'discount' flag to be 0 in the BOOKORDER table. If this is not true, then the discount is not applied.

In order to fulfil the trigger several new columns have been added to the existing BOOK and BOOKORDER tables.
To BOOK:

- BOOK_STOCK defaulting to 50 for the testing of the trigger. In a business setting this would be updated when inventory was received.
- BOOK_STOCK_MIN defaulting to 15 for testing the trigger. Likewise, in a business setting this would be identified over time and set differently for each item.
- BOOK_REORDER default to 0. A Boolean flag that identifies if a book needs reordering. The following business process to identify the amount worth reordering and lodging the reorder is not covered.

To BOOKORDER:

- BO_COST added to hold the order total cost calculated as an incremental sum with each item added to the order.
- BO_DISCOUNT as a Boolean flag default to 0. Identified is the discount has been applied to the order and control the logic to not add further, unprofitable discounts.

Code:

```
/*ALTER TABLE CODE FOR TRIGGER*/
ALTER TABLE BOOKORDER
ADD BO_COST NUMBER(7,2)
DEFAULT 0
;

ALTER TABLE BOOKORDER
ADD BO_DISCOUNT NUMBER(1)
DEFAULT 0
;

ALTER TABLE BOOKORDER
ADD CONSTRAINT DISCOUNT_FLAG CHECK (bo_discount IN (0, 1))
;

ALTER TABLE BOOK
ADD BOOK_STOCK NUMBER(4)
DEFAULT 50
;

ALTER TABLE BOOK
ADD BOOK_STOCK_MIN NUMBER(2)
DEFAULT 15
;

ALTER TABLE BOOK
ADD BOOK_REORDER NUMBER(1)
DEFAULT 0
;

ALTER TABLE BOOK
ADD CONSTRAINT REORDER_FLAG CHECK (book_reorder IN (0, 1))
;

--------------------------------------------------------------------------------
```

```
/*TRIGGER CODE*/
CREATE OR REPLACE TRIGGER total_order_trg_2
AFTER INSERT ON BOOKORDERITEM

FOR EACH ROW

DECLARE
    total NUMBER(7,2) := 0;
    ordernum NUMBER(4);
    qty NUMBER(2) := 0;
    stockredux NUMBER(2) :=0;
    stockmin NUMBER(2) :=0;
    stock_level_error EXCEPTION;
    discount NUMBER(1) :=0;

BEGIN

    --1)get bo_order
    SELECT bo_ordernum INTO ordernum
    FROM BOOKORDER
    WHERE BOOKORDER.bo_ordernum = :new.bo_ordernum;

    --2) update bookorder total cost
    SELECT
        :new.boi_qty * book_retail INTO total
    FROM BOOK
    WHERE book_isbn = :new.book_isbn;

    --3) update bookorder cost
    UPDATE BOOKORDER
    SET BOOKORDER.bo_cost = BOOKORDER.bo_cost + total
    WHERE BOOKORDER.bo_ordernum = :new.bo_ordernum;


    --4) reduce book stock
    qty := :new.boi_qty;

    SELECT
        BOOK.book_stock_min INTO stockmin
    FROM BOOK
    WHERE BOOK.book_isbn = :new.book_isbn;

    SELECT
        BOOK.book_stock - qty INTO stockredux
    FROM BOOK
    WHERE BOOK.book_isbn = :new.book_isbn;

    UPDATE BOOK
```

```
    SET BOOK.book_stock = BOOK.book_stock - qty
    WHERE BOOK.book_isbn = :new.book_isbn;

    /*--DEBUG code
    DBMS_OUTPUT.PUT_LINE(ordernum);
    DBMS_OUTPUT.PUT_LINE(qty);
    DBMS_OUTPUT.PUT_LINE(stockredux);
    DBMS_OUTPUT.PUT_LINE(total);
    */

    --5) check stock level
    IF stockredux < 0
    THEN
        DBMS_OUTPUT.PUT_LINE ('NOT ENOUGH STOCK');
        RAISE stock_level_error;

        ELSE IF stockredux < stockmin
        THEN
            DBMS_OUTPUT.PUT_LINE ('reorder ticket raised');
            UPDATE BOOK
            SET BOOK.book_reorder = 1
            WHERE BOOK.book_isbn = :new.book_isbn;
        END IF;
    END IF;

    --6) Check and apply 10% discount to order over $200
    SELECT
        bo_cost INTO total
    FROM BOOKORDER
    WHERE BOOKORDER.bo_ordernum = :new.bo_ordernum;

    SELECT
        BO_discount INTO discount
     FROM BOOKORDER
    WHERE BOOKORDER.bo_ordernum = :new.bo_ordernum;

    IF total > 200 AND discount = 0
    THEN total := total * 0.9;
        UPDATE BOOKORDER
        SET BOOKORDER.bo_cost = total,
            BOOKORDER.bo_discount = 1
        WHERE BOOKORDER.bo_ordernum = :new.bo_ordernum;
        DBMS_OUTPUT.PUT_LINE('Discount Applied, purchase reduced to');
        DBMS_OUTPUT.PUT_LINE(total);

        ELSE IF total > 200 AND discount = 1
        THEN
            UPDATE BOOKORDER
```

```
            SET BOOKORDER.bo_cost = total
            WHERE BOOKORDER.bo_ordernum = :new.bo_ordernum;
        END IF;
    END IF;

EXCEPTION
    WHEN stock_level_error
    THEN raise_application_error(-20002, 'Not Enough Stock for sale');

END;
/
--------------------------------------------------------------------------------
```

Test:

```
/* TEST CODE */
INSERT INTO  BOOKORDERITEM
VALUES (9999, 1, '0000000001', 5); --now have 45 stock, not discounted

INSERT INTO  BOOKORDERITEM
VALUES (9999, 2, '0000000001', 15); -- now have 30 stock, discounted

INSERT INTO  BOOKORDERITEM
VALUES (9999, 3, '0000000001', 10); -- now have 20 stock, not discounted

INSERT INTO  BOOKORDERITEM
VALUES (9999, 4, '0000000001', 10); --trigger reorder, now have 10 stock, not
discounted

INSERT INTO  BOOKORDERITEM
VALUES (9999, 5, '0000000001', 10); --now have 0 stock, not discounted

INSERT INTO  BOOKORDERITEM
VALUES (9999, 6, '0000000001', 10); --trigger out of stock exception

INSERT INTO  BOOKORDERITEM
VALUES (9999, 7, '0401140733', 5); --add another book to the order, no
discount

INSERT INTO  BOOKORDERITEM
VALUES (1000, 9, '3437212490', 2); -- order with no discount, stock reduced to
48

INSERT INTO  BOOKORDERITEM
VALUES (1000, 10, '3437212490', 50); -- stock would go below 0, trigger out of
stock exception
/*--------------------------------------------------------------------------*/
```

Output:

Before the trigger:

BOOK: all stock levels at 50, all stock minimums at 15 and all reorders set to 0.

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY	BOOK_ST...	BOOK_STOCK_MIN	BOOK_REO...
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS	50	15	0
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS	50	15	0
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE	50	15	0
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN	50	15	0
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER	50	15	0
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING	50	15	1
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER	50	15	0
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER	50	15	0
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER	50	15	0
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE	50	15	0
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING	50	15	0
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN	50	15	0
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP	50	15	0
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS	50	15	0
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE	50	15	0
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION	50	15	0

BOOKORDER: all costs set to 0, all discount flags at 0.

	BO_ORDERNUM	CUST_NUM	BO_ORDERDATE	BO_SHIPDATE	BO_SHIPSTREET	BO_SHIPCITY	BO_SHIPSTATE	BO_SHIPZIP	BO_COST	BO_DISCOUNT
1	9999	1001	12/10/23	12/10/23	test	test	TT	12345	0	0
2	1000	1005	31/03/23	02/04/23	1201 ORANGE AVE	SEATTLE	WA	98114	0	0
3	1001	1010	31/03/23	01/04/23	114 EAST SAVANNAH	ATLANTA	GA	30314	0	0
4	1002	1011	31/03/23	01/04/23	58 TILA CIRCLE	CHICAGO	IL	60605	0	0
5	1003	1001	01/04/23	01/04/23	958 MAGNOLIA LANE	EASTPOINT	FL	32328	0	0
6	1004	1020	01/04/23	05/04/23	561 ROUNDABOUT WAY	TRENTON	NJ	08601	0	0
7	1005	1018	01/04/23	02/04/23	1008 GRAND AVENUE	MACON	GA	31206	0	0
8	1006	1003	01/04/23	02/04/23	558A CAPITOL HWY.	TALLAHASSEE	FL	32307	0	0
9	1007	1007	02/04/23	04/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
10	1008	1004	02/04/23	03/04/23	69821 SOUTH AVENUE	BOISE	ID	83707	0	0
11	1009	1005	03/04/23	05/04/23	9 LIGHTENING RD.	SEATTLE	WA	98110	0	0
12	1010	1019	03/04/23	04/04/23	384 WRONG WAY HOME	MORRISTOWN	NJ	07960	0	0
13	1011	1010	03/04/23	05/04/23	102 WEST LAFAYETTE	ATLANTA	GA	30311	0	0
14	1012	1017	03/04/23	(null)	1295 WINDY AVENUE	KALMAZOO	MI	49002	0	0
15	1013	1014	03/04/23	04/04/23	7618 MOUNTAIN RD.	CODY	WY	82414	0	0
16	1014	1007	04/04/23	05/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
17	1015	1020	04/04/23	(null)	557 GLITTER ST.	TRENTON	NJ	08606	0	0
18	1016	1003	04/04/23	(null)	9901 SEMINOLE WAY	TALLAHASSEE	FL	32307	0	0
19	1017	1015	04/04/23	05/04/23	887 HOT ASPHALT ST	MIAMI	FL	33112	0	0
20	1018	1001	05/04/23	(null)	95812 HIGHWAY 98	EASTPOINT	FL	32328	0	0
21	1019	1018	05/04/23	(null)	1008 GRAND AVENUE	MACON	GA	31206	0	0
22	1020	1008	05/04/23	(null)	195 JAMISON LANE	CHEYENNE	WY	82003	0	0

BOOKORDERITEM has no record for order number 9999

	BO_ORDERNUM	BOI_ITEMNUM	BOOK_ISBN	BOI_QTY
1	1000	1	3437212490	1
2	1001	1	9247381001	1
3	1001	2	2491748320	1
4	1002	1	8843172113	2
5	1003	1	8843172113	1
6	1003	2	1059831198	1
7	1003	3	3437212490	1
8	1004	1	2491748320	2
9	1005	1	2147428890	1
10	1006	1	9959789321	1
11	1007	1	3957136468	3
12	1007	2	9959789321	1
13	1007	3	8117949391	1
14	1007	4	8843172113	1
15	1008	1	3437212490	2
16	1009	1	3437212490	1
17	1009	2	0401140733	1
18	1010	1	8843172113	1
19	1011	1	2491748320	1
20	1012	1	8117949391	1
21	1012	2	1915762492	2
22	1012	3	2491748320	1
23	1012	4	0401140733	1
24	1013	1	8843172113	1
25	1014	1	0401140733	2
26	1015	1	3437212490	1
27	1016	1	2491748320	1
28	1017	1	8117949391	2
29	1018	1	3437212490	1
30	1018	2	8843172113	1
31	1019	1	0401140733	1
32	1020	1	3437212490	1

After the trigger:

First trigger. BOOKORDERITEM has record; 9999, 1 with 5 items. BOOKORDER has increased to $115.9 and BOOK stock for 'The Brothers Karamazov' has reduced to 45.

BOOKORDER ITEM:

	BO_ORDERNUM	BOI_ITEMNUM	BOOK_ISBN	BOI_QTY
1	9999	1	0000000001	5
2	1000	1	3437212490	1
3	1001	1	9247381001	1
4	1001	2	2491748320	1
5	1002	1	8843172113	2
6	1003	1	8843172113	1
7	1003	2	1059831198	1
8	1003	3	3437212490	1
9	1004	1	2491748320	2
10	1005	1	2147428890	1
11	1006	1	9959789321	1
12	1007	1	3957136468	3
13	1007	2	9959789321	1
14	1007	3	8117949391	1
15	1007	4	8843172113	1
16	1008	1	3437212490	2
17	1009	1	3437212490	1
18	1009	2	0401140733	1
19	1010	1	8843172113	1
20	1011	1	2491748320	1
21	1012	1	8117949391	1
22	1012	2	1915762492	2
23	1012	3	2491748320	1
24	1012	4	0401140733	1
25	1013	1	8843172113	1
26	1014	1	0401140733	2
27	1015	1	3437212490	1
28	1016	1	2491748320	1
29	1017	1	8117949391	2
30	1018	1	3437212490	1
31	1018	2	8843172113	1
32	1019	1	0401140733	1
33	1020	1	3437212490	1

BOOK:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY	BOOK_ST...	BOOK_STOCK_MIN	BOOK_REO...
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS	50	15	0
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS	50	15	0
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE	50	15	0
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN	50	15	0
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER	50	15	0
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING	50	15	0
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER	50	15	0
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER	50	15	0
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER	50	15	0
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE	50	15	0
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING	50	15	0
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN	50	15	0
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP	50	15	0
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS	50	15	0
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE	50	15	0
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION	45	15	0

BOOKORDER:

	BO_ORDERNUM	CUST_NUM	BO_ORDERDATE	BO_SHIPDATE	BO_SHIPSTREET	BO_SHIPCITY	BO_SHIPSTATE	BO_SHIPZIP	BO_COST	BO_DISCOUNT
1	9999	1001	12/10/23	12/10/23	test	test	TT	12345	115.9	0
2	1000	1005	31/03/23	02/04/23	1201 ORANGE AVE	SEATTLE	WA	98114	0	0
3	1001	1010	31/03/23	01/04/23	114 EAST SAVANNAH	ATLANTA	GA	30314	0	0
4	1002	1011	31/03/23	01/04/23	58 TILA CIRCLE	CHICAGO	IL	60605	0	0
5	1003	1001	01/04/23	01/04/23	958 MAGNOLIA LANE	EASTPOINT	FL	32328	0	0
6	1004	1020	01/04/23	05/04/23	561 ROUNDABOUT WAY	TRENTON	NJ	08601	0	0
7	1005	1018	01/04/23	02/04/23	1008 GRAND AVENUE	MACON	GA	31206	0	0
8	1006	1003	01/04/23	02/04/23	558A CAPITOL HWY.	TALLAHASSEE	FL	32307	0	0
9	1007	1007	02/04/23	04/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
10	1008	1004	02/04/23	03/04/23	69821 SOUTH AVENUE	BOISE	ID	83707	0	0
11	1009	1005	03/04/23	05/04/23	9 LIGHTENING RD.	SEATTLE	WA	98110	0	0
12	1010	1019	03/04/23	04/04/23	384 WRONG WAY HOME	MORRISTOWN	NJ	07960	0	0
13	1011	1010	03/04/23	05/04/23	102 WEST LAFAYETTE	ATLANTA	GA	30311	0	0
14	1012	1017	03/04/23	(null)	1295 WINDY AVENUE	KALMAZOO	MI	49002	0	0
15	1013	1014	03/04/23	04/04/23	7618 MOUNTAIN RD.	CODY	WY	82414	0	0
16	1014	1007	04/04/23	05/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
17	1015	1020	04/04/23	(null)	557 GLITTER ST.	TRENTON	NJ	08606	0	0
18	1016	1003	04/04/23	(null)	9901 SEMINOLE WAY	TALLAHASSEE	FL	32307	0	0
19	1017	1015	04/04/23	05/04/23	887 HOT ASPHALT ST	MIAMI	FL	33112	0	0
20	1018	1001	05/04/23	(null)	95812 HIGHWAY 98	EASTPOINT	FL	32328	0	0
21	1019	1018	05/04/23	(null)	1008 GRAND AVENUE	MACON	GA	31206	0	0
22	1020	1008	05/04/23	(null)	195 JAMISON LANE	CHEYENNE	WY	82003	0	0

By 23021700, 23023649, 20001249

Second trigger. BOOKORDERITEM has record; 9999, 2 with 15 items. BOOKORDER increases, discount is activated, cost has increased to $417.24 and BOOK stock for 'The Brothers Karamazov' has reduced to 30. System message d

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Discount Applied,purchase reduced to
417.24
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BOOKORDER ITEM:

	BO_ORDERNUM	BOI_ITEMNUM	BOOK_ISBN	BOI_QTY
1	9999	1	0000000001	5
2	9999	2	0000000001	15
3	1000	1	3437212490	1
4	1001	1	9247381001	1
5	1001	2	2491748320	1
6	1002	1	8843172113	2
7	1003	1	8843172113	1
8	1003	2	1059831198	1
9	1003	3	3437212490	1
10	1004	1	2491748320	2
11	1005	1	2147428890	1
12	1006	1	9959789321	1
13	1007	1	3957136468	3
14	1007	2	9959789321	1
15	1007	3	8117949391	1
16	1007	4	8843172113	1
17	1008	1	3437212490	2
18	1009	1	3437212490	1
19	1009	2	0401140733	1
20	1010	1	8843172113	1
21	1011	1	2491748320	1
22	1012	1	8117949391	1
23	1012	2	1915762492	2
24	1012	3	2491748320	1
25	1012	4	0401140733	1
26	1013	1	8843172113	1
27	1014	1	0401140733	2
28	1015	1	3437212490	1
29	1016	1	2491748320	1
30	1017	1	8117949391	2
31	1018	1	3437212490	1
32	1018	2	8843172113	1
33	1019	1	0401140733	1
34	1020	1	3437212490	1

BOOK:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY	BOOK_ST...	BOOK_STOCK_MIN	BOOK_REO...
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS	50	15	0
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS	50	15	0
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE	50	15	0
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN	50	15	0
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER	50	15	0
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING	50	15	0
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER	50	15	0
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER	50	15	0
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER	50	15	0
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE	50	15	0
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING	50	15	0
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN	50	15	0
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP	50	15	0
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS	50	15	0
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE	50	15	0
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION	30	15	0

BOOKORDER:

	BO_ORDE...	CUST_NUM	BO_ORDERDATE	BO_SHIPDATE	BO_SHIPSTREET	BO_SHIPCITY	BO_SHIPSTATE	BO_SHIPZIP	BO_COST	BO_DISCOUNT
1	9999	1001	12/10/23	12/10/23	test	test	TT	12345	417.24	1
2	1000	1005	31/03/23	02/04/23	1201 ORANGE AVE	SEATTLE	WA	98114	0	0
3	1001	1010	31/03/23	01/04/23	114 EAST SAVANNAH	ATLANTA	GA	30314	0	0
4	1002	1011	31/03/23	01/04/23	58 TILA CIRCLE	CHICAGO	IL	60605	0	0
5	1003	1001	01/04/23	01/04/23	958 MAGNOLIA LANE	EASTPOINT	FL	32328	0	0
6	1004	1020	01/04/23	05/04/23	561 ROUNDABOUT WAY	TRENTON	NJ	08601	0	0
7	1005	1018	01/04/23	02/04/23	1008 GRAND AVENUE	MACON	GA	31206	0	0
8	1006	1003	01/04/23	02/04/23	558A CAPITOL HWY.	TALLAHASSEE	FL	32307	0	0
9	1007	1007	02/04/23	04/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
10	1008	1004	02/04/23	03/04/23	69821 SOUTH AVENUE	BOISE	ID	83707	0	0
11	1009	1005	03/04/23	05/04/23	9 LIGHTENING RD.	SEATTLE	WA	98110	0	0
12	1010	1019	03/04/23	04/04/23	384 WRONG WAY HOME	MORRISTOWN	NJ	07960	0	0
13	1011	1010	03/04/23	05/04/23	102 WEST LAFAYETTE	ATLANTA	GA	30311	0	0
14	1012	1017	03/04/23	(null)	1295 WINDY AVENUE	KALMAZOO	MI	49002	0	0
15	1013	1014	03/04/23	04/04/23	7618 MOUNTAIN RD.	CODY	WY	82414	0	0
16	1014	1007	04/04/23	05/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
17	1015	1020	04/04/23	(null)	557 GLITTER ST.	TRENTON	NJ	08606	0	0
18	1016	1003	04/04/23	(null)	9901 SEMINOLE WAY	TALLAHASSEE	FL	32307	0	0
19	1017	1015	04/04/23	05/04/23	887 HOT ASPHALT ST	MIAMI	FL	33112	0	0
20	1018	1001	05/04/23	(null)	95812 HIGHWAY 98	EASTPOINT	FL	32328	0	0
21	1019	1018	05/04/23	(null)	1008 GRAND AVENUE	MACON	GA	31206	0	0
22	1020	1008	05/04/23	(null)	195 JAMISON LANE	CHEYENNE	WY	82003	0	0

Third trigger. BOOKORDERITEM has record; 9999, 3 with 10 items. BOOKORDER increases, to $649.04, no discount and BOOK stock for 'The Brothers Karamazov' has reduced to 20.

BOOKORDER ITEM:

	BO_ORDERNUM	BOI_ITEMNUM	BOOK_ISBN	BOI_QTY
1	9999	1	0000000001	5
2	9999	2	0000000001	15
3	9999	3	0000000001	10
4	1000	1	3437212490	1
5	1001	1	9247381001	1
6	1001	2	2491748320	1
7	1002	1	8843172113	2
8	1003	1	8843172113	1
9	1003	2	1059831198	1
10	1003	3	3437212490	1
11	1004	1	2491748320	2
12	1005	1	2147428890	1
13	1006	1	9959789321	1
14	1007	1	3957136468	3
15	1007	2	9959789321	1
16	1007	3	8117949391	1
17	1007	4	8843172113	1
18	1008	1	3437212490	2
19	1009	1	3437212490	1
20	1009	2	0401140733	1
21	1010	1	8843172113	1
22	1011	1	2491748320	1
23	1012	1	8117949391	1
24	1012	2	1915762492	2
25	1012	3	2491748320	1
26	1012	4	0401140733	1
27	1013	1	8843172113	1
28	1014	1	0401140733	2
29	1015	1	3437212490	1
30	1016	1	2491748320	1
31	1017	1	8117949391	2
32	1018	1	3437212490	1
33	1018	2	8843172113	1
34	1019	1	0401140733	1
35	1020	1	3437212490	1

BOOK:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY	BOOK_ST...	BOOK_STOCK_MIN	BOOK_REO...
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS	50	15	0
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS	50	15	0
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE	50	15	0
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN	50	15	0
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER	50	15	0
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING	50	15	0
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER	50	15	0
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER	50	15	0
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER	50	15	0
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE	50	15	0
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING	50	15	0
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN	50	15	0
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP	50	15	0
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS	50	15	0
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE	50	15	0
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION	20	15	0

BOOKORDER:

	BO_ORDERNUM	CUST_NUM	BO_ORDERDATE	BO_SHIPDATE	BO_SHIPSTREET	BO_SHIPCITY	BO_SHIPSTATE	BO_SHIPZIP	BO_COST	BO_DISCOUNT
1	9999	1001	12/10/23	12/10/23	test	test	TT	12345	649.04	1
2	1000	1005	31/03/23	02/04/23	1201 ORANGE AVE	SEATTLE	WA	98114	0	0
3	1001	1010	31/03/23	01/04/23	114 EAST SAVANNAH	ATLANTA	GA	30314	0	0
4	1002	1011	31/03/23	01/04/23	58 TILA CIRCLE	CHICAGO	IL	60605	0	0
5	1003	1001	01/04/23	01/04/23	958 MAGNOLIA LANE	EASTPOINT	FL	32328	0	0
6	1004	1020	01/04/23	05/04/23	561 ROUNDABOUT WAY	TRENTON	NJ	08601	0	0
7	1005	1018	01/04/23	02/04/23	1008 GRAND AVENUE	MACON	GA	31206	0	0
8	1006	1003	01/04/23	02/04/23	558A CAPITOL HWY.	TALLAHASSEE	FL	32307	0	0
9	1007	1007	02/04/23	04/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
10	1008	1004	02/04/23	03/04/23	69821 SOUTH AVENUE	BOISE	ID	83707	0	0
11	1009	1005	03/04/23	05/04/23	9 LIGHTENING RD.	SEATTLE	WA	98110	0	0
12	1010	1019	03/04/23	04/04/23	384 WRONG WAY HOME	MORRISTOWN	NJ	07960	0	0
13	1011	1010	03/04/23	05/04/23	102 WEST LAFAYETTE	ATLANTA	GA	30311	0	0
14	1012	1017	03/04/23	(null)	1295 WINDY AVENUE	KALMAZOO	MI	49002	0	0
15	1013	1014	03/04/23	04/04/23	7618 MOUNTAIN RD.	CODY	WY	82414	0	0
16	1014	1007	04/04/23	05/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
17	1015	1020	04/04/23	(null)	557 GLITTER ST.	TRENTON	NJ	08606	0	0
18	1016	1003	04/04/23	(null)	9901 SEMINOLE WAY	TALLAHASSEE	FL	32307	0	0
19	1017	1015	04/04/23	05/04/23	887 HOT ASPHALT ST	MIAMI	FL	33112	0	0
20	1018	1001	05/04/23	(null)	95812 HIGHWAY 98	EASTPOINT	FL	32328	0	0
21	1019	1018	05/04/23	(null)	1008 GRAND AVENUE	MACON	GA	31206	0	0
22	1020	1008	05/04/23	(null)	195 JAMISON LANE	CHEYENNE	WY	82003	0	0

By 23021700, 23023649, 20001249

Fourth trigger. BOOKORDERITEM has record; 9999, 4 with 10 items. BOOKORDER increases, to $880.84 and BOOK stock for 'The Brothers Karamazov' has reduced to 10, reorder set to 1, system `reorder ticket raised`

BOOKORDER ITEM:

	BO_ORDERNUM	BOI_ITEMNUM	BOOK_ISBN	BOI_QTY
1	9999	4	0000000001	10
2	9999	1	0000000001	5
3	9999	2	0000000001	15
4	9999	3	0000000001	10
5	1000	1	3437212490	1
6	1001	1	9247381001	1
7	1001	2	2491748320	1
8	1002	1	8843172113	2
9	1003	1	8843172113	1
10	1003	2	1059831198	1
11	1003	3	3437212490	1
12	1004	1	2491748320	2
13	1005	1	2147428890	1
14	1006	1	9959789321	1
15	1007	1	3957136468	3
16	1007	2	9959789321	1
17	1007	3	8117949391	1
18	1007	4	8843172113	1
19	1008	1	3437212490	2
20	1009	1	3437212490	1
21	1009	2	0401140733	1
22	1010	1	8843172113	1
23	1011	1	2491748320	1
24	1012	1	8117949391	1
25	1012	2	1915762492	2
26	1012	3	2491748320	1
27	1012	4	0401140733	1
28	1013	1	8843172113	1
29	1014	1	0401140733	2
30	1015	1	3437212490	1
31	1016	1	2491748320	1
32	1017	1	8117949391	2
33	1018	1	3437212490	1
34	1018	2	8843172113	1
35	1019	1	0401140733	1
36	1020	1	3437212490	1

BOOK:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY	BOOK_ST...	BOOK_STOCK_MIN	BOOK_REO...
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS	50	15	0
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS	50	15	0
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE	50	15	0
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN	50	15	0
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER	50	15	0
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING	50	15	0
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER	50	15	0
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER	50	15	0
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER	50	15	0
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE	50	15	0
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING	50	15	0
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN	50	15	0
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP	50	15	0
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS	50	15	0
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE	50	15	0
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION	10	15	1

BOOKORDER:

	BO_ORDERNUM	CUST_NUM	BO_ORDERDATE	BO_SHIPDATE	BO_SHIPSTREET	BO_SHIPCITY	BO_SHIPSTATE	BO_SHIPZIP	BO_COST	BO_DISCOUNT
1	9999	1001	12/10/23	12/10/23	test	test	TT	12345	880.84	1
2	1000	1005	31/03/23	02/04/23	1201 ORANGE AVE	SEATTLE	WA	98114	0	0
3	1001	1010	31/03/23	01/04/23	114 EAST SAVANNAH	ATLANTA	GA	30314	0	0
4	1002	1011	31/03/23	01/04/23	58 TILA CIRCLE	CHICAGO	IL	60605	0	0
5	1003	1001	01/04/23	01/04/23	958 MAGNOLIA LANE	EASTPOINT	FL	32328	0	0
6	1004	1020	01/04/23	05/04/23	561 ROUNDABOUT WAY	TRENTON	NJ	08601	0	0
7	1005	1018	01/04/23	02/04/23	1008 GRAND AVENUE	MACON	GA	31206	0	0
8	1006	1003	01/04/23	02/04/23	558A CAPITOL HWY.	TALLAHASSEE	FL	32307	0	0
9	1007	1007	02/04/23	04/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
10	1008	1004	02/04/23	03/04/23	69821 SOUTH AVENUE	BOISE	ID	83707	0	0
11	1009	1005	03/04/23	05/04/23	9 LIGHTENING RD.	SEATTLE	WA	98110	0	0
12	1010	1019	03/04/23	04/04/23	384 WRONG WAY HOME	MORRISTOWN	NJ	07960	0	0
13	1011	1010	03/04/23	05/04/23	102 WEST LAFAYETTE	ATLANTA	GA	30311	0	0
14	1012	1017	03/04/23	(null)	1295 WINDY AVENUE	KALMAZOO	MI	49002	0	0
15	1013	1014	03/04/23	04/04/23	7618 MOUNTAIN RD.	CODY	WY	82414	0	0
16	1014	1007	04/04/23	05/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
17	1015	1020	04/04/23	(null)	557 GLITTER ST.	TRENTON	NJ	08606	0	0
18	1016	1003	04/04/23	(null)	9901 SEMINOLE WAY	TALLAHASSEE	FL	32307	0	0
19	1017	1015	04/04/23	05/04/23	887 HOT ASPHALT ST	MIAMI	FL	33112	0	0
20	1018	1001	05/04/23	(null)	95812 HIGHWAY 98	EASTPOINT	FL	32328	0	0
21	1019	1018	05/04/23	(null)	1008 GRAND AVENUE	MACON	GA	31206	0	0
22	1020	1008	05/04/23	(null)	195 JAMISON LANE	CHEYENNE	WY	82003	0	0

By 23021700, 23023649, 20001249

Fifth trigger. BOOKORDERITEM has record; 9999, 5 with 10 items. BOOKORDER increases, to $1112.64 and BOOK stock for 'The Brothers Karamazov' has reduced to 0.

BOOKORDER ITEM:

	BO_ORDERNUM	BOI_ITEMNUM	BOOK_ISBN	BOI_QTY
1	9999	4	0000000001	10
2	9999	1	0000000001	5
3	9999	2	0000000001	15
4	9999	5	0000000001	10
5	9999	3	0000000001	10
6	1000	1	3437212490	1
7	1001	1	9247381001	1
8	1001	2	2491748320	1
9	1002	1	8843172113	2
10	1003	1	8843172113	1
11	1003	2	1059831198	1
12	1003	3	3437212490	1
13	1004	1	2491748320	2
14	1005	1	2147428890	1
15	1006	1	9959789321	1
16	1007	1	3957136468	3
17	1007	2	9959789321	1
18	1007	3	8117949391	1
19	1007	4	8843172113	1
20	1008	1	3437212490	2
21	1009	1	3437212490	1
22	1009	2	0401140733	1
23	1010	1	8843172113	1
24	1011	1	2491748320	1
25	1012	1	8117949391	1
26	1012	2	1915762492	2
27	1012	3	2491748320	1
28	1012	4	0401140733	1
29	1013	1	8843172113	1
30	1014	1	0401140733	2
31	1015	1	3437212490	1
32	1016	1	2491748320	1
33	1017	1	8117949391	2
34	1018	1	3437212490	1
35	1018	2	8843172113	1
36	1019	1	0401140733	1
37	1020	1	3437212490	1

BOOK:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY	BOOK_ST...	BOOK_STOCK_MIN	BOOK_REO...
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS	50	15	0
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS	50	15	0
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE	50	15	0
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN	50	15	0
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER	50	15	0
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING	50	15	0
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER	50	15	0
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER	50	15	0
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER	50	15	0
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE	50	15	0
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING	50	15	0
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN	50	15	0
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP	50	15	0
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS	50	15	0
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE	50	15	0
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION	0	15	1

BOOKORDER:

	BO_ORDERNUM	CUST_NUM	BO_ORDERDATE	BO_SHIPDATE	BO_SHIPSTREET	BO_SHIPCITY	BO_SHIPSTATE	BO_SHIPZIP	BO_COST	BO_DISCOUNT
1	9999	1001	12/10/23	12/10/23	test	test	TT	12345	1112.64	1
2	1000	1005	31/03/23	02/04/23	1201 ORANGE AVE	SEATTLE	WA	98114	0	0
3	1001	1010	31/03/23	01/04/23	114 EAST SAVANNAH	ATLANTA	GA	30314	0	0
4	1002	1011	31/03/23	01/04/23	58 TILA CIRCLE	CHICAGO	IL	60605	0	0
5	1003	1001	01/04/23	01/04/23	958 MAGNOLIA LANE	EASTPOINT	FL	32328	0	0
6	1004	1020	01/04/23	05/04/23	561 ROUNDABOUT WAY	TRENTON	NJ	08601	0	0
7	1005	1018	01/04/23	02/04/23	1008 GRAND AVENUE	MACON	GA	31206	0	0
8	1006	1003	01/04/23	02/04/23	558A CAPITOL HWY.	TALLAHASSEE	FL	32307	0	0
9	1007	1007	02/04/23	04/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
10	1008	1004	02/04/23	03/04/23	69821 SOUTH AVENUE	BOISE	ID	83707	0	0
11	1009	1005	03/04/23	05/04/23	9 LIGHTENING RD.	SEATTLE	WA	98110	0	0
12	1010	1019	03/04/23	04/04/23	384 WRONG WAY HOME	MORRISTOWN	NJ	07960	0	0
13	1011	1010	03/04/23	05/04/23	102 WEST LAFAYETTE	ATLANTA	GA	30311	0	0
14	1012	1017	03/04/23	(null)	1295 WINDY AVENUE	KALMAZOO	MI	49002	0	0
15	1013	1014	03/04/23	04/04/23	7618 MOUNTAIN RD.	CODY	WY	82414	0	0
16	1014	1007	04/04/23	05/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
17	1015	1020	04/04/23	(null)	557 GLITTER ST.	TRENTON	NJ	08606	0	0
18	1016	1003	04/04/23	(null)	9901 SEMINOLE WAY	TALLAHASSEE	FL	32307	0	0
19	1017	1015	04/04/23	05/04/23	887 HOT ASPHALT ST	MIAMI	FL	33112	0	0
20	1018	1001	05/04/23	(null)	95812 HIGHWAY 98	EASTPOINT	FL	32328	0	0
21	1019	1018	05/04/23	(null)	1008 GRAND AVENUE	MACON	GA	31206	0	0
22	1020	1008	05/04/23	(null)	195 JAMISON LANE	CHEYENNE	WY	82003	0	0

By 23021700, 23023649, 20001249

Sixth trigger. BOOKORDERITEM has record; 9999, 6 with 10 items **not** created. System throws exception message.

```
Error starting at line : 165 in command -
INSERT INTO  BOOKORDERITEM
VALUES (9999, 6, '0000000001', 10)
Error report -
ORA-20002: Not Enough Stock for sale
ORA-06512: at "GROUP35.TOTAL_ORDER_TRG_2", line 98
ORA-04088: error during execution of trigger 'GROUP35.TOTAL_ORDER_TRG_2'
```

Seventh trigger. BOOKORDERITEM has record; 9999, 7 with 5 items of another book. BOOKORDER increases, to $1222.64 and BOOK stock for 'Revenge of Mickey' has reduced to 45.

BOOKORDER ITEM:

	BO_ORDERNUM	BOI_ITEMNUM	BOOK_ISBN	BOI_QTY
1	9999	4	0000000001	10
2	9999	1	0000000001	5
3	9999	2	0000000001	15
4	9999	5	0000000001	10
5	9999	7	0401140733	5
6	9999	3	0000000001	10
7	1000	1	3437212490	1
8	1001	1	9247381001	1
9	1001	2	2491748320	1
10	1002	1	8843172113	2
11	1003	1	8843172113	1
12	1003	2	1059831198	1
13	1003	3	3437212490	1
14	1004	1	2491748320	2
15	1005	1	2147428890	1
16	1006	1	9959789321	1
17	1007	1	3957136468	3
18	1007	2	9959789321	1
19	1007	3	8117949391	1
20	1007	4	8843172113	1
21	1008	1	3437212490	2
22	1009	1	3437212490	1
23	1009	2	0401140733	1
24	1010	1	8843172113	1
25	1011	1	2491748320	1
26	1012	1	8117949391	1
27	1012	2	1915762492	2
28	1012	3	2491748320	1
29	1012	4	0401140733	1
30	1013	1	8843172113	1
31	1014	1	0401140733	2
32	1015	1	3437212490	1
33	1016	1	2491748320	1
34	1017	1	8117949391	2
35	1018	1	3437212490	1
36	1018	2	8843172113	1
37	1019	1	0401140733	1
38	1020	1	3437212490	1

BOOK:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY	BOOK_ST...	BOOK_STOCK_MIN	BOOK_REO...
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS	50	15	0
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS	50	15	0
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE	45	15	0
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN	50	15	0
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER	50	15	0
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING	50	15	0
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER	50	15	0
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER	50	15	0
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER	50	15	0
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE	50	15	0
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING	50	15	0
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN	50	15	0
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP	50	15	0
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS	50	15	0
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE	50	15	0
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION	0	15	1

BOOKORDER:

	BO_ORDERNUM	CUST_NUM	BO_ORDERDATE	BO_SHIPDATE	BO_SHIPSTREET	BO_SHIPCITY	BO_SHIPSTATE	BO_SHIPZIP	BO_COST	BO_DISCOUNT
1	9999	1001	12/10/23	12/10/23	test	test	TT	12345	1222.64	1
2	1000	1005	31/03/23	02/04/23	1201 ORANGE AVE	SEATTLE	WA	98114	0	0
3	1001	1010	31/03/23	01/04/23	114 EAST SAVANNAH	ATLANTA	GA	30314	0	0
4	1002	1011	31/03/23	01/04/23	58 TILA CIRCLE	CHICAGO	IL	60605	0	0
5	1003	1001	01/04/23	01/04/23	958 MAGNOLIA LANE	EASTPOINT	FL	32328	0	0
6	1004	1020	01/04/23	05/04/23	561 ROUNDABOUT WAY	TRENTON	NJ	08601	0	0
7	1005	1018	01/04/23	02/04/23	1008 GRAND AVENUE	MACON	GA	31206	0	0
8	1006	1003	01/04/23	02/04/23	558A CAPITOL HWY.	TALLAHASSEE	FL	32307	0	0
9	1007	1007	02/04/23	04/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
10	1008	1004	02/04/23	03/04/23	69821 SOUTH AVENUE	BOISE	ID	83707	0	0
11	1009	1005	03/04/23	05/04/23	9 LIGHTENING RD.	SEATTLE	WA	98110	0	0
12	1010	1019	03/04/23	04/04/23	384 WRONG WAY HOME	MORRISTOWN	NJ	07960	0	0
13	1011	1010	03/04/23	05/04/23	102 WEST LAFAYETTE	ATLANTA	GA	30311	0	0
14	1012	1017	03/04/23	(null)	1295 WINDY AVENUE	KALMAZOO	MI	49002	0	0
15	1013	1014	03/04/23	04/04/23	7618 MOUNTAIN RD.	CODY	WY	82414	0	0
16	1014	1007	04/04/23	05/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
17	1015	1020	04/04/23	(null)	557 GLITTER ST.	TRENTON	NJ	08606	0	0
18	1016	1003	04/04/23	(null)	9901 SEMINOLE WAY	TALLAHASSEE	FL	32307	0	0
19	1017	1015	04/04/23	05/04/23	887 HOT ASPHALT ST	MIAMI	FL	33112	0	0
20	1018	1001	05/04/23	(null)	95812 HIGHWAY 98	EASTPOINT	FL	32328	0	0
21	1019	1018	05/04/23	(null)	1008 GRAND AVENUE	MACON	GA	31206	0	0
22	1020	1008	05/04/23	(null)	195 JAMISON LANE	CHEYENNE	WY	82003	0	0

By 23021700, 23023649, 20001249

Eighth trigger. BOOKORDERITEM has record; 1000, 9 with 2 added. BOOKORDER increases, to $39.9 and BOOK stock for 'Cooking with mushrooms' has reduced to 48.

BOOKORDER ITEM:

	BO_ORDERNUM	BOI_ITEMNUM	BOOK_ISBN	BOI_QTY
1	9999	4	0000000001	10
2	9999	1	0000000001	5
3	9999	2	0000000001	15
4	9999	5	0000000001	10
5	9999	7	0401140733	5
6	9999	3	0000000001	10
7	1000	9	3437212490	2
8	1000	1	3437212490	1
9	1001	1	9247381001	1
10	1001	2	2491748320	1
11	1002	1	8843172113	2
12	1003	1	8843172113	1
13	1003	2	1059831198	1
14	1003	3	3437212490	1
15	1004	1	2491748320	2
16	1005	1	2147428890	1
17	1006	1	9959789321	1
18	1007	1	3957136468	3
19	1007	2	9959789321	1
20	1007	3	8117949391	1
21	1007	4	8843172113	1
22	1008	1	3437212490	2
23	1009	1	3437212490	1
24	1009	2	0401140733	1
25	1010	1	8843172113	1
26	1011	1	2491748320	1
27	1012	1	8117949391	1
28	1012	2	1915762492	2
29	1012	3	2491748320	1
30	1012	4	0401140733	1
31	1013	1	8843172113	1
32	1014	1	0401140733	2
33	1015	1	3437212490	1
34	1016	1	2491748320	1
35	1017	1	8117949391	2
36	1018	1	3437212490	1
37	1018	2	8843172113	1
38	1019	1	0401140733	1
39	1020	1	3437212490	1

BOOK:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY	BOOK_ST...	BOOK_STOCK_MIN	BOOK_REO...
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS	50	15	0
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS	50	15	0
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE	45	15	0
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN	50	15	0
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER	50	15	0
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING	48	15	0
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER	50	15	0
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER	50	15	0
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER	50	15	0
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE	50	15	0
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING	50	15	0
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN	50	15	0
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP	50	15	0
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS	50	15	0
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE	50	15	0
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION	0	15	1

BOOKORDER:

	BO_ORDERNUM	CUST_NUM	BO_ORDERDATE	BO_SHIPDATE	BO_SHIPSTREET	BO_SHIPCITY	BO_SHIPSTATE	BO_SHIPZIP	BO_COST	BO_DISCOUNT
1	9999	1001	12/10/23	12/10/23	test	test	TT	12345	1222.64	1
2	1000	1005	31/03/23	02/04/23	1201 ORANGE AVE	SEATTLE	WA	98114	39.9	0
3	1001	1010	31/03/23	01/04/23	114 EAST SAVANNAH	ATLANTA	GA	30314	0	0
4	1002	1011	31/03/23	01/04/23	58 TILA CIRCLE	CHICAGO	IL	60605	0	0
5	1003	1001	01/04/23	01/04/23	958 MAGNOLIA LANE	EASTPOINT	FL	32328	0	0
6	1004	1020	01/04/23	05/04/23	561 ROUNDABOUT WAY	TRENTON	NJ	08601	0	0
7	1005	1018	01/04/23	02/04/23	1008 GRAND AVENUE	MACON	GA	31206	0	0
8	1006	1003	01/04/23	02/04/23	558A CAPITOL HWY.	TALLAHASSEE	FL	32307	0	0
9	1007	1007	02/04/23	04/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
10	1008	1004	02/04/23	03/04/23	69821 SOUTH AVENUE	BOISE	ID	83707	0	0
11	1009	1005	03/04/23	05/04/23	9 LIGHTENING RD.	SEATTLE	WA	98110	0	0
12	1010	1019	03/04/23	04/04/23	384 WRONG WAY HOME	MORRISTOWN	NJ	07960	0	0
13	1011	1010	03/04/23	05/04/23	102 WEST LAFAYETTE	ATLANTA	GA	30311	0	0
14	1012	1017	03/04/23	(null)	1295 WINDY AVENUE	KALMAZOO	MI	49002	0	0
15	1013	1014	03/04/23	04/04/23	7618 MOUNTAIN RD.	CODY	WY	82414	0	0
16	1014	1007	04/04/23	05/04/23	9153 MAIN STREET	AUSTIN	TX	78710	0	0
17	1015	1020	04/04/23	(null)	557 GLITTER ST.	TRENTON	NJ	08606	0	0
18	1016	1003	04/04/23	(null)	9901 SEMINOLE WAY	TALLAHASSEE	FL	32307	0	0
19	1017	1015	04/04/23	05/04/23	887 HOT ASPHALT ST	MIAMI	FL	33112	0	0
20	1018	1001	05/04/23	(null)	95812 HIGHWAY 98	EASTPOINT	FL	32328	0	0
21	1019	1018	05/04/23	(null)	1008 GRAND AVENUE	MACON	GA	31206	0	0
22	1020	1008	05/04/23	(null)	195 JAMISON LANE	CHEYENNE	WY	82003	0	0

By 23021700, 23023649, 20001249

Ninth trigger. BOOKORDERITEM has record; 1000, 10 with 50 NOT added. Not enough stock error message displayed.

```
Error starting at line : 174 in command -
INSERT INTO  BOOKORDERITEM
VALUES (1000, 10, '3437212490', 50)
Error report -
ORA-20002: Not Enough Stock for sale
ORA-06512: at "GROUP35.TOTAL_ORDER_TRG_2", line 98
ORA-04088: error during execution of trigger 'GROUP35.TOTAL_ORDER_TRG_2'
```

Question Bb)

Write a trigger that does not allow the book retail price to be updated when the increase (in retail price) is over 25%. Provide test data and corresponding results to confirm that the trigger works.

Code:

```
CREATE TRIGGER retail_control AFTER UPDATE OF book_retail
ON BOOK
FOR EACH ROW

DECLARE
    retail_control_error EXCEPTION;

BEGIN
    IF :new.book_retail/:old.book_retail > 1.25
    THEN DBMS_OUTPUT.PUT_LINE ('Cannot raise retail price by more than 25%');
    RAISE retail_control_error;
    END IF;

EXCEPTION
    WHEN retail_control_error
    THEN raise_application_error(-20001, 'Cannot update');
END;
/
```

Test:

Negative Test:

```
UPDATE BOOK
    SET book_retail = 400
WHERE book_isbn = 8117949391
;
```

Positive test:

```
UPDATE BOOK
    SET book_retail = 10
WHERE book_isbn = 8117949391
;
```

Output:

Negative test:

```
Error starting at line : 68 in command -
UPDATE BOOK
    SET book_retail = 400
WHERE book_isbn = 8117949391
Error report -
ORA-20001: Cannot update
ORA-06512: at "GROUP35.RETAIL_CONTROL", line 12
ORA-04088: error during execution of trigger 'GROUP35.RETAIL_CONTROL'
```

Positive test:

Before:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS
2	1234795860	Procedure Book Test	09/10/23	1	20	23.18	COMPUTER
3	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS
4	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE
5	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN
6	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER
7	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING
8	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER
9	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER
10	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER
11	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE
12	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING
13	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN
14	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP
15	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS
16	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE

After:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS
2	1234795860	Procedure Book Test	09/10/23	1	20	23.18	COMPUTER
3	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS
4	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE
5	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN
6	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER
7	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING
8	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER
9	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER
10	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER
11	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE
12	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING
13	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	10	CHILDREN
14	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP
15	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS
16	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE

Question Bc)

Write a trigger that does not allow more than three author names to be associated with books under FITNESS category (e.g., if a Book is added, it should only allow up to 3 book authors to be recorded in BookAuthor table for 'fitness' category books). Provide the appropriate test data and results.

Code:

```
CREATE OR REPLACE TRIGGER trg_fitness_author_control
FOR INSERT ON BOOKAUTHOR
COMPOUND TRIGGER
  TYPE type_isbns  IS TABLE OF BOOK.book_isbn%TYPE;
  v_isbns          type_isbns := type_isbns();
  v_book_category  BOOK.book_category%TYPE;
  v_count          INTEGER;

  AFTER EACH ROW IS
  BEGIN
    v_isbns.EXTEND;
    v_isbns(v_isbns.COUNT) := :new.book_isbn;
  END AFTER EACH ROW;

  AFTER STATEMENT IS
  BEGIN
    FOR i IN 1 .. v_isbns.COUNT LOOP
      SELECT B.book_category, a.cnt
      INTO v_book_category, v_count
      FROM BOOK B
      CROSS APPLY
      (
        SELECT COUNT(*) as cnt
        FROM BOOKAUTHOR BA
        WHERE BA.book_isbn = B.book_isbn
      ) A
      WHERE B.book_isbn = v_isbns(i);

      IF v_book_category = 'FITNESS' AND v_count > 3 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Too many authors for fitness book '
|| v_isbns(i));
      END IF;
    END LOOP;
  END AFTER STATEMENT;
END trg_fitness_author_control;
/
```

Test:

```
--inserting test data into book, author and book author

INSERT INTO BOOK VALUES ('9000000099', 'FITEST', '21-FEB-23', 01, 10.00,
20.95, 'FITNESS', 1, 20.95);

INSERT INTO AUTHOR VALUES ('M200', 'Mycenae', 'Agamemnon');
INSERT INTO AUTHOR VALUES ('I200', 'Ithica', 'Odysseus');
INSERT INTO AUTHOR VALUES ('D200', 'Dardania', 'Aeneas');

INSERT INTO BOOKAUTHOR VALUES ('9000000099', 'M200');
INSERT INTO BOOKAUTHOR VALUES ('9000000099', 'I200');
INSERT INTO BOOKAUTHOR VALUES ('9000000099', 'D200');
```

```
--test by adding one more author to a fitness book with 3 already
INSERT INTO BOOKAUTHOR VALUES ('9000000099', 'P105');
```

Output:

```
Error starting at line : 139 in command -
INSERT INTO BOOKAUTHOR VALUES ('9000000099', 'P105')
Error report -
ORA-20001: Too many authors for fitness book 9000000099
ORA-06512: at "GROUP35.TRG_FITNESS_AUTHOR_CONTROL", line 28
ORA-04088: error during execution of trigger 'GROUP35.TRG_FITNESS_AUTHOR_CONTROL'
```

Question Bd)

Write a procedure to insert a new book record. The procedure should also automatically calculate the book retail value. This retail is calculated as 110.5% of the book cost price plus 4.5% of the average cost price of the existing books in the database. Provide rest of the attributes' values as input parameters. Execute your procedure to insert at least one book record.

Code:

Note: This code assumes that publishers are added via a different business process because they require contact and phone details. There are no checks on the insert for duplicates as these are handled by constraints already existing on the book, book author and author tables.

```
CREATE OR REPLACE PROCEDURE PRC_INSERT_NEW_BOOK_RECORD2(
    p_Book_ISBN IN BOOK.Book_ISBN%TYPE,
    p_Book_Title IN BOOK.Book_Title%TYPE,
    p_Book_PubDate IN BOOK.Book_PubDate%TYPE,
    p_Pub_ID IN BOOK.Pub_ID%TYPE,
    p_Book_Cost IN BOOK.Book_Cost%TYPE,
    p_Book_Category IN BOOK.Book_Category%TYPE,
    p_Author_ID IN AUTHOR.Author_id%TYPE,
    p_Author_LName IN AUTHOR.Author_LName%TYPE,
    p_Author_FName IN AUTHOR.Author_FName%TYPE
    )

IS
    v_Avg_Cost BOOK.Book_Cost%TYPE;
    v_Book_Retail BOOK.Book_Retail%TYPE;
BEGIN
    SELECT AVG(Book_Cost) INTO v_Avg_Cost FROM BOOK;
    v_Book_Retail := ((1.105 * p_Book_Cost) + (0.045 * v_Avg_Cost));

    INSERT INTO BOOK ( Book_ISBN, Book_Title, Book_PubDate, Pub_ID, Book_Cost,
Book_Retail, Book_Category)
    VALUES ( p_Book_ISBN, p_Book_Title, p_Book_PubDate, p_Pub_ID, p_Book_Cost,
v_Book_Retail, p_Book_Category);
    INSERT INTO AUTHOR (Author_id, Author_Lname, Author_Fname)
    VALUES (p_Author_ID, p_Author_LName, p_Author_FName);
    INSERT INTO BOOKAUTHOR (Book_isbn, Author_id)
    VALUES (p_Book_ISBN, p_Author_ID);
    COMMIT;
    DBMS_OUTPUT.PUT_LINE('New book record inserted successfully.');

END;
/
```

Test:

```
-- Test insert of book record
```

```
EXECUTE PRC_INSERT_NEW_BOOK_RECORD2('0000000001', 'THE BROTHERS
KARAMAZOV','09/10/23', 1, 20.00, 'FICTION',0,0, 'Z999', 'Fyodor', 'Dostoy');
```

Output:

Before:

Book:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE

Author:

	AUTHOR_ID	AUTHOR_LNAME	AUTHOR_FNAME
1	M200	Mycenae	Agamemnon
2	I200	Ithica	Odysseus
3	D200	Dardania	Aeneas
4	S100	SMITH	SAM
5	J100	JONES	JANICE
6	A100	AUSTIN	JAMES
7	M100	MARTINEZ	SHEILA
8	K100	KZOCHSKY	TAMARA
9	P100	PORTER	LISA
10	A105	ADAMS	JUAN
11	B100	BAKER	JACK
12	P105	PETERSON	TINA
13	W100	WHITE	WILLIAM
14	W105	WHITE	LISA
15	R100	ROBINSON	ROBERT
16	F100	FIELDS	OSCAR
17	W110	WILKINSON	ANTHONY

Book Author:

	BOOK_ISBN	AUTHOR_
1	9000000099	M200
2	9000000099	I200
3	9000000099	D200
4	1059831198	S100
5	1059831198	P100
6	0401140733	J100
7	4981341710	K100
8	8843172113	P105
9	8843172113	A100
10	8843172113	A105
11	3437212490	B100
12	3957136468	A100
13	1915762492	W100
14	1915762492	W105
15	9959789321	J100
16	2491748320	R100
17	2491748320	F100
18	2491748320	B100
19	0299282519	S100
20	8117949391	R100
21	0132149871	S100
22	9247381001	W100
23	2147428890	W105

After:

Book:

	BOOK_ISBN	BOOK_TITLE	BOOK_PUBDATE	PUB_ID	BOOK_COST	BOOK_RETAIL	BOOK_CATEGORY
1	9000000099	FITEST	21/02/23	1	10	20.95	FITNESS
2	1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/01	4	18.75	30.95	FITNESS
3	0401140733	REVENGE OF MICKEY	14/12/01	1	14.2	22	FAMILY LIFE
4	4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/02	2	37.8	59.95	CHILDREN
5	8843172113	DATABASE IMPLEMENTATION	04/06/99	3	31.4	55.95	COMPUTER
6	3437212490	COOKING WITH MUSHROOMS	28/02/00	4	12.5	19.95	COOKING
7	3957136468	HOLY GRAIL OF ORACLE	31/12/01	3	47.25	75.95	COMPUTER
8	1915762492	HANDCRANKED COMPUTERS	21/01/01	3	21.8	25	COMPUTER
9	9959789321	E-BUSINESS THE EASY WAY	01/03/02	2	37.9	54.5	COMPUTER
10	2491748320	PAINLESS CHILD-REARING	17/07/00	5	48	89.95	FAMILY LIFE
11	0299282519	THE WOK WAY TO COOK	11/09/00	4	19	28.75	COOKING
12	8117949391	BIG BEAR AND LITTLE DOVE	08/11/01	5	5.32	8.95	CHILDREN
13	0132149871	HOW TO GET FASTER PIZZA	11/11/02	4	17.85	29.95	SELF HELP
14	9247381001	HOW TO MANAGE THE MANAGER	09/05/99	1	15.4	31.95	BUSINESS
15	2147428890	SHORTEST POEMS	01/05/01	5	21.85	39.95	LITERATURE
16	0000000001	THE BROTHERS KARAMAZOV	09/10/23	1	20	23.18	FICTION

Author:

	AUTHOR_ID	AUTHOR_LNAME	AUTHOR_FNAME
1	M200	Mycenae	Agamemnon
2	I200	Ithica	Odysseus
3	D200	Dardania	Aeneas
4	S100	SMITH	SAM
5	J100	JONES	JANICE
6	A100	AUSTIN	JAMES
7	M100	MARTINEZ	SHEILA
8	K100	KZOCHSKY	TAMARA
9	P100	PORTER	LISA
10	A105	ADAMS	JUAN
11	B100	BAKER	JACK
12	P105	PETERSON	TINA
13	W100	WHITE	WILLIAM
14	W105	WHITE	LISA
15	R100	ROBINSON	ROBERT
16	F100	FIELDS	OSCAR
17	W110	WILKINSON	ANTHONY
18	Z999	Fyodor	Dostoy

Book Author:

	BOOK_ISBN	AUTHOR_ID
1	9000000099	M200
2	9000000099	I200
3	9000000099	D200
4	1059831198	S100
5	1059831198	P100
6	0401140733	J100
7	4981341710	K100
8	8843172113	P105
9	8843172113	A100
10	8843172113	A105
11	3437212490	B100
12	3957136468	A100
13	1915762492	W100
14	1915762492	W105
15	9959789321	J100
16	2491748320	R100
17	2491748320	F100
18	2491748320	B100
19	0299282519	S100
20	8117949391	R100
21	0132149871	S100
22	9247381001	W100
23	2147428890	W105
24	0000000001	Z999

Inserting an existing record

```
Error starting at line : 162 in command -
BEGIN PRC_INSERT_NEW_BOOK_RECORD('0000000001', 'THE BROTHERS KARAMAZOV','09/10/23', 1, 20.00, 'FICTION','Z999', 'Fyodor', 'Dostoy'); END;
Error report -
ORA-00001: unique constraint (GROUP35.SYS_C00262184) violated
ORA-06512: at "GROUP35.PRC_INSERT_NEW_BOOK_RECORD", line 23
ORA-06512: at line 1
00001. 00000 -  "unique constraint (%s.%s) violated"
*Cause:    An UPDATE or INSERT statement attempted to insert a duplicate key.
           For Trusted Oracle configured in DBMS MAC mode, you may see
           this message if a duplicate entry exists at a different level.
*Action:   Either remove the unique restriction or do not insert the key.
```

Question Be)

Use a cursor construct to list book authors for the 'computer' category books (along with their book title, cost, and retail). Present your work in a well readable report format (e.g., include headings, well-spaced expressions, some summary in the end).

Code:

```
DECLARE CURSOR computer_books_cursor IS

SELECT
    B.Book_title,
    B.Book_cost,
    B.Book_retail,
    LISTAGG( A.Author_Fname || ' ' || A.Author_LName, ', ') AS Author_Name
FROM BOOK B
JOIN BOOKAUTHOR BA ON B.Book_isbn = BA.book_isbn
JOIN AUTHOR A ON BA.Author_id = A.author_id
WHERE B.Book_Category = 'COMPUTER'
GROUP BY B.Book_title, B.Book_cost, B.Book_retail, B.book_category;

book_info computer_books_cursor%ROWTYPE;

BEGIN
    DBMS_OUTPUT.PUT_LINE('-------------------------------------START OF
REPORT-------------------------------------');
    DBMS_OUTPUT.PUT_LINE('Book Title               |Book Cost |Book
Retail|Author Name                              |');
    DBMS_OUTPUT.PUT_LINE('-------------------------------------------------------
-------------------------------------');

OPEN computer_books_cursor;
LOOP
    FETCH computer_books_cursor INTO book_info;
    IF computer_books_cursor%FOUND THEN
        DBMS_OUTPUT.PUT_LINE(RPAD(book_info.Book_Title, 24) || ' | ' ||
                        TO_CHAR(book_info.Book_Cost, 9999.99) || ' | ' ||
                        TO_CHAR(book_info.Book_Retail, 9999.99) || ' | ' ||
                        RPAD(book_info.Author_Name, 40)  || ' | '
);
    ELSE
        EXIT;
    END IF;

END LOOP;
    DBMS_OUTPUT.PUT_LINE('-------------------------------------------------------
-------------------------------------');
    DBMS_OUTPUT.PUT_LINE('Total number of rows    ' ||
computer_books_cursor%ROWCOUNT);
```

```
    DBMS_OUTPUT.PUT_LINE('----------------------------------------END OF
REPORT----------------------------------------');
      IF computer_books_cursor%ISOPEN THEN
            CLOSE computer_books_cursor;
      END IF;

END;
/
```

Output:

```
------------------------------------START OF REPORT-----------------------------------
Book Title               |Book Cost |Book Retail|Author Name                              |
-------------------------------------------------------------------------------------------
DATABASE IMPLEMENTATION  |    31.40 |    55.95 | JAMES AUSTIN, JUAN ADAMS, TINA PETERSON  |
E-BUSINESS THE EASY WAY  |    37.90 |    54.50 | JANICE JONES                             |
HANDCRANKED COMPUTERS    |    21.80 |    25.00 | WILLIAM WHITE, LISA WHITE                |
HOLY GRAIL OF ORACLE     |    47.25 |    75.95 | JAMES AUSTIN                             |
-------------------------------------------------------------------------------------------
Total number of rows    4
----------------------------------------END OF REPORT----------------------------------


PL/SQL procedure successfully completed.
```

Question Bf)

Write a function to appropriately format money values (e.g., columns such as book cost, retail price uses a currency sign, commas if needed) when presented in some listing. Call this function in a SQL statement for displaying books' costs and retail prices.

Code:

```
CREATE OR REPLACE FUNCTION FUNC_MONEY_FORMAT(v_amount NUMBER)
RETURN VARCHAR2 IS
v_formatted_amount VARCHAR2(100);
BEGIN
  v_formatted_amount := TO_CHAR(v_amount, '$999,999,999.99');
RETURN v_formatted_amount;

 DBMS_OUTPUT.PUT_LINE(FUNC_MONEY_FORMAT(1000000));
END;
/
```

Test:

```
--Test Code
SELECT
    Book_Title AS "Book Title",
    FUNC_MONEY_FORMAT(Book_Cost) AS "Book Cost",
    FUNC_MONEY_FORMAT(Book_Retail) AS "Book Retail"
FROM
    Book
;
```

Output:

Book Title	Book Cost	Book Retail
FITEST	$10.00	$20.95
Procedure Book Test	$20.00	$23.18
BODYBUILD IN 10 MINUTES A DAY	$18.75	$30.95
REVENGE OF MICKEY	$14.20	$22.00
BUILDING A CAR WITH TOOTHPICKS	$37.80	$59.95
DATABASE IMPLEMENTATION	$31.40	$55.95
COOKING WITH MUSHROOMS	$12.50	$19.95
HOLY GRAIL OF ORACLE	$47.25	$75.95
HANDCRANKED COMPUTERS	$21.80	$25.00
E-BUSINESS THE EASY WAY	$37.90	$54.50
PAINLESS CHILD-REARING	$48.00	$89.95
THE WOK WAY TO COOK	$19.00	$28.75
BIG BEAR AND LITTLE DOVE	$5.32	$8.95
HOW TO GET FASTER PIZZA	$17.85	$29.95
HOW TO MANAGE THE MANAGER	$15.40	$31.95
SHORTEST POEMS	$21.85	$39.95

	Book Title	Book Cost	Book Retail
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3	BODYBUILD IN 10 MINUTES A DAY	$18.75	$30.95
4	REVENGE OF MICKEY	$14.20	$22.00
5	BUILDING A CAR WITH TOOTHPICKS	$37.80	$59.95
6	DATABASE IMPLEMENTATION	$31.40	$55.95
7	COOKING WITH MUSHROOMS	$12.50	$19.95
8	HOLY GRAIL OF ORACLE	$47.25	$75.95
9	HANDCRANKED COMPUTERS	$21.80	$25.00
10	E-BUSINESS THE EASY WAY	$37.90	$54.50
11	PAINLESS CHILD-REARING	$48.00	$89.95
12	THE WOK WAY TO COOK	$19.00	$28.75
13	BIG BEAR AND LITTLE DOVE	$5.32	$8.95
14	HOW TO GET FASTER PIZZA	$17.85	$29.95
15	HOW TO MANAGE THE MANAGER	$15.40	$31.95
16	SHORTEST POEMS	$21.85	$39.95

Section C (LINQ)

Question Ca)

List books with the retail price $25 or over.

Code:

```
var question1 =
    from s in Books
    where s.BookRetail > 25
    select s;
question1.Dump();
```

Output:

BookIsbn	BookTitle	BookPubdate	PubID	BookCost	BookRetail	BookCategory	Bookorderitems	Bookauthors
1059831198	BODYBUILDIN 10 MINUTES A DAY	21/01/2001 0:00	4	18.75	30.95	FITNESS	Bookorderitems	Bookauthors
4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/2002 0:00	2	37.8	59.95	CHILDREN	Bookorderitems	Bookauthors
8843172113	DATABASE IMPLEMENTATION	4/06/1999 0:00	3	31.4	55.95	COMPUTER	Bookorderitems	Bookauthors
3957136468	HOLY GRAIL OF ORACLE	31/12/2001 0:00	3	47.25	75.95	COMPUTER	Bookorderitems	Bookauthors
9959789321	E-BUSINESS THE EASY WAY	1/03/2002 0:00	2	37.9	54.5	COMPUTER	Bookorderitems	Bookauthors
2491748320	PAINLESS CHILD-REARING	17/07/2000 0:00	5	48	89.95	FAMILY LIFE	Bookorderitems	Bookauthors
299282519	THE WOK WAY TO COOK	11/09/2000 0:00	4	19	28.75	COOKING	Bookorderitems	Bookauthors
132149871	HOW TO GET FASTER PIZZA	11/11/2002 0:00	4	17.85	29.95	SELF HELP	Bookorderitems	Bookauthors
9247381001	HOW TO MANAGE THE MANAGER	9/05/1999 0:00	1	15.4	31.95	BUSINESS	Bookorderitems	Bookauthors
2147428890	SHORTEST POEMS	1/05/2001 0:00	5	21.85	39.95	LITERATURE	Bookorderitems	Bookauthors
				295.2	497.85			

Query<Book> (10 items)

BookIsbn	BookTitle	BookPubdate	PubID	BookCost	BookRetail	BookCategory	Bookorderitems	Bookauthors
1059831198	BODYBUILD IN 10 MINUTES A DAY	21/01/2001 12:00:00 am	4	18.75	30.95	FITNESS	Bookorderitems	Bookauthors
4981341710	BUILDING A CAR WITH TOOTHPICKS	18/03/2002 12:00:00 am	2	37.8	59.95	CHILDREN	Bookorderitems	Bookauthors
8843172113	DATABASE IMPLEMENTATION	4/06/1999 12:00:00 am	3	31.4	55.95	COMPUTER	Bookorderitems	Bookauthors
3957136468	HOLY GRAIL OF ORACLE	31/12/2001 12:00:00 am	3	47.25	75.95	COMPUTER	Bookorderitems	Bookauthors
9959789321	E-BUSINESS THE EASY WAY	1/03/2002 12:00:00 am	2	37.9	54.5	COMPUTER	Bookorderitems	Bookauthors
2491748320	PAINLESS CHILD-REARING	17/07/2000 12:00:00 am	5	48	89.95	FAMILY LIFE	Bookorderitems	Bookauthors
0299282519	THE WOK WAY TO COOK	11/09/2000 12:00:00 am	4	19	28.75	COOKING	Bookorderitems	Bookauthors
0132149871	HOW TO GET FASTER PIZZA	11/11/2002 12:00:00 am	4	17.85	29.95	SELF HELP	Bookorderitems	Bookauthors
9247381001	HOW TO MANAGE THE MANAGER	9/05/1999 12:00:00 am	1	15.4	31.95	BUSINESS	Bookorderitems	Bookauthors
2147428890	SHORTEST POEMS	1/05/2001 12:00:00 am	5	21.85	39.95	LITERATURE	Bookorderitems	Bookauthors
				295.20	497.85			

Question Cb)

List books that have the word “HOW” in the book title.

Code:

```
var question2 =
    from s in Books
    where s.BookTitle.Contains("HOW")
    select s;
question2.Dump();
```

Output:

BookIsbn	BookTitle	BookPubdate	PubID	BookCost	BookRetail	BookCategory	Bookorderitems	Bookauthors
132149871	HOW TO GET FASTER PIZZA	11/11/2002 0:00	4	17.85	29.95	SELF HELP	Bookorderitems	Bookauthors
9247381001	HOW TO MANAGE THE MANAGER	9/05/1999 0:00	1	15.4	31.95	BUSINESS	Bookorderitems	Bookauthors
				33.25	61.9			

Query<Book> (2 items)

BookIsbn	BookTitle	BookPubdate	PubID	BookCost	BookRetail	BookCategory	Bookorderitems	Bookauthors
0132149871	HOW TO GET FASTER PIZZA	11/11/2002 12:00:00 am	4	17.85	29.95	SELF HELP	Bookorderitems	Bookauthors
9247381001	HOW TO MANAGE THE MANAGER	9/05/1999 12:00:00 am	1	15.4	31.95	BUSINESS	Bookorderitems	Bookauthors
				33.25	61.90			

Question Cc)

List book categories, book titles and publisher names.

Code:

```
var question3  =
    from s in Books
    join p in Publishers
        on s.PubID equals p.PubID
    select new {s.BookTitle, s.BookCategory, p.PubName};
question3.Dump();
```

Output:

BookTitle	BookCategory	PubName
REVENGE OF MICKEY	FAMILY LIFE	PRINTING IS US
HOW TO MANAGE THE MANAGER	BUSINESS	PRINTING IS US
E-BUSINESS THE EASY WAY	COMPUTER	PUBLISH OUR WAY
BUILDING A CAR WITH TOOTHPICKS	CHILDREN	PUBLISH OUR WAY
HOLY GRAIL OF ORACLE	COMPUTER	AMERICAN PUBLISHING
DATABASE IMPLEMENTATION	COMPUTER	AMERICAN PUBLISHING
HANDCRANKED COMPUTERS	COMPUTER	AMERICAN PUBLISHING
COOKING WITH MUSHROOMS	COOKING	READING MATERIALS INC.
THE WOK WAY TO COOK	COOKING	READING MATERIALS INC.
HOW TO GET FASTER PIZZA	SELF HELP	READING MATERIALS INC.
BODYBUILD IN 10 MINUTES A DAY	FITNESS	READING MATERIALS INC.
PAINLESS CHILD-REARING	FAMILY LIFE	REED-N-RITE
BIG BEAR AND LITTLE DOVE	CHILDREN	REED-N-RITE
SHORTEST POEMS	LITERATURE	REED-N-RITE

Query<> (14 items)

BookTitle	BookCategory	PubName
REVENGE OF MICKEY	FAMILY LIFE	PRINTING IS US
HOW TO MANAGE THE MANAGER	BUSINESS	PRINTING IS US
E-BUSINESS THE EASY WAY	COMPUTER	PUBLISH OUR WAY
BUILDING A CAR WITH TOOTHPICKS	CHILDREN	PUBLISH OUR WAY
HOLY GRAIL OF ORACLE	COMPUTER	AMERICAN PUBLISHING
DATABASE IMPLEMENTATION	COMPUTER	AMERICAN PUBLISHING
HANDCRANKED COMPUTERS	COMPUTER	AMERICAN PUBLISHING
COOKING WITH MUSHROOMS	COOKING	READING MATERIALS INC.
THE WOK WAY TO COOK	COOKING	READING MATERIALS INC.
HOW TO GET FASTER PIZZA	SELF HELP	READING MATERIALS INC.
BODYBUILD IN 10 MINUTES A DAY	FITNESS	READING MATERIALS INC.
PAINLESS CHILD-REARING	FAMILY LIFE	REED-N-RITE
BIG BEAR AND LITTLE DOVE	CHILDREN	REED-N-RITE
SHORTEST POEMS	LITERATURE	REED-N-RITE

Question Cd)

Display total number of books by each publisher in the order of publisher ID.

Code:

```
var question4 =
    from b in Books
    join p in Publishers
        on b.PubID equals p.PubID
    group b by b.PubID into newgroup
    orderby newgroup.Key
    select new {
        PublisherID = newgroup.Key,
        TotalBooks = newgroup.Count()
        } into g
    join p in Publishers
        on g.PublisherID equals p.PubID
    select new{
        PublisherID = g.PublisherID,
        Publisher = p.PubName,
        TotalBooks = g.TotalBooks};
question4.Dump();
```

Output:

PublisherID	Publisher	TotalBooks
1	PRINTING IS US	2
2	PUBLISH OUR WAY	2
3	AMERICAN PUBLISHING	3
4	READING MATERIALS INC.	4
5	REED-N-RITE	3
		14

Query<> (5 items)

PublisherID	Publisher	TotalBooks
1	PRINTING IS US	2
2	PUBLISH OUR WAY	2
3	AMERICAN PUBLISHING	3
4	READING MATERIALS INC.	4
5	REED-N-RITE	3
		14

Question Ce)

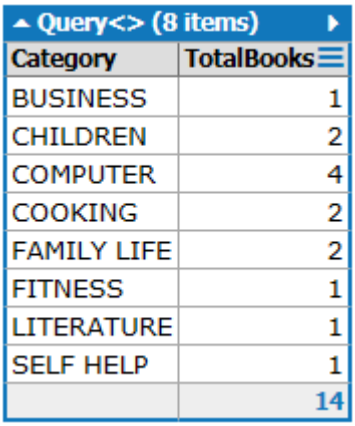
Display the number of books under each category in the order of category.

Code:

```
var question5 =
    from b in Books
    group b by b.BookCategory into newgroup
    orderby newgroup.Key
    select new {
        Category = newgroup.Key,
        TotaBooks = newgroup.Count()
        };
question5.Dump();
```

Output:

Category	TotalBooks
BUSINESS	1
CHILDREN	2
COMPUTER	4
COOKING	2
FAMILY LIFE	2
FITNESS	1
LITERATURE	1
SELF HELP	1
	14



Query<> (8 items)

Category	TotalBooks
BUSINESS	1
CHILDREN	2
COMPUTER	4
COOKING	2
FAMILY LIFE	2
FITNESS	1
LITERATURE	1
SELF HELP	1
	14

Section D (NOSQL / MONGODB)

Question Da)

Setting up the database of hobbits

Code:

```
//Insert the data

db.hobbits.insert({name: "Bilbo Baggins", dob: new Date(1992,2,13,7,47),
loves:["horse riding","surfing"],weight: 45, gender: "m", cash: 3000});

db.hobbits.insert({name: "Rosie-Posie", dob: new Date(1961, 0, 24, 13, 0),
loves: ["horse riding", "chess"], weight: 63, gender: "f", cash: 2250});

db.hobbits.insert({name: "Bingo Deepdelver", dob: new Date(1963, 1, 9, 22,
10), loves: ["scrabble", "basketball"], weight: 72, gender: "m", cash: 9000});

db.hobbits.insert({name: "Samwise Deepdelver", dob: new Date(1990, 7, 18, 18,
44), loves: ["cricket"],weight: 69, gender: "m", cash: 8000});

db.hobbits.insert({name: "Dora Toadfoot", dob: new Date(1985, 6, 4, 2, 1),
loves:["cricket", "horse riding", "trekking"], weight:60, gender:"f",
cash:6000});

db.hobbits.insert({name:"Kennoy Dippy", dob: new Date(1997, 6, 1, 10, 42),
loves: ["chess", "rugby"],weight: 69, gender: "m", cash: 500});

db.hobbits.insert({name: "Rajaram Ghoda", dob: new Date(1985, 4, 3, 0, 57),
loves: ["cricket", "boxing"],weight: 101, gender: "m", cash: 6500});

db.hobbits.insert({name: "Dolly Boffin", dob: new Date(1968, 9, 8, 14, 53),
loves: ["cricket", "dancing"], weight: 52, gender: "f", cash: 6000});

db.hobbits.insert({name: "Ayna Shyna", dob: new Date(1958, 2, 7, 8, 30),
loves: ["singing", "dancing"], weight: 73, gender: "f", cash: 5000});

db.hobbits.insert({name: "Lynx Todka", dob: new Date(1960, 2, 7, 8, 30),
loves: ["singing", "dancing","horse riding"], weight: 93, gender: "m", cash:
5500});

db.hobbits.insert({name: "Aria Lyaria", dob: new Date(1964, 1, 9, 8, 40),
loves: ["chess", "snooker","horse riding"], weight: 83, gender: "f", cash:
6500});

db.hobbits.insert({name: "Tane Shane", dob: new Date(1990, 3, 7, 9, 30),
loves: ["singing", "dancing","horse riding"], weight: 98, gender: "m", cash:
5500});
```

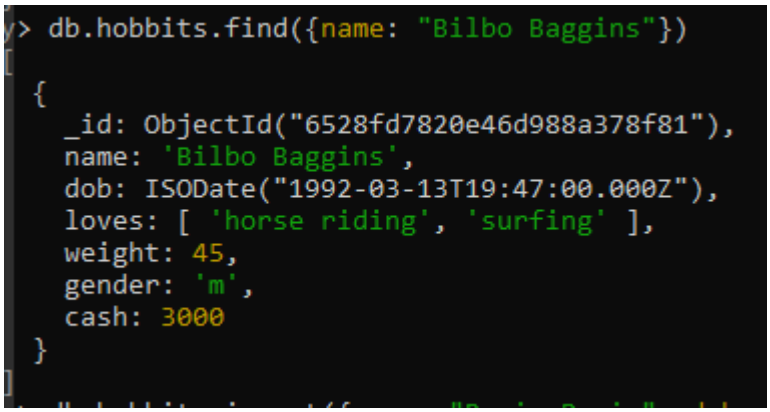
```
db.hobbits.insert({name: "Tui Shui", dob: new Date(1975, 2, 8, 10, 30), loves:
["singing", "dancing","rugby"], weight: 75, gender: "f", cash: 7800});

db.hobbits.insert({name: "Tamati Shamati", dob: new Date(1991, 1, 7, 7, 30),
loves: ["rugby", "soccer","go carting"], weight: 93, gender: "m", cash:
9500});

db.hobbits.insert({name: "Sheroo Beroo", dob: new Date(1999, 7, 7, 8, 25),
loves: ["kite flying", "kabaadi","cricket", "stapu"], weight: 88, gender: "m",
cash: 7500});
```

Output:

Finding one of the hobbits



Question Db)

Generating a gender wise average weight report

Code:

```
//create the mapping output
var mapSectionD = function() {
    emit(this.gender, { totalWeight: this.weight, count: 1 });
};

//calculate the average weight by sex
var reduceSectionD = function(key, values) {
    var reduced = { totalWeight: 0, count: 0, avgWeight: 0 };

    values.forEach(function(value) {
        reduced.totalWeight += value.totalWeight;
        reduced.count += value.count;
    });

    if (reduced.count > 0) {
        reduced.avgWeight = reduced.totalWeight / reduced.count;
    }

    return reduced;
};

//specify the output
db.hobbits.mapReduce(
    mapSectionD,
    reduceSectionD,
    {
        out: "hobbits_SectionD"
    }
);

//call the output
db.hobbits_SectionD.find();
```

Output:

```
test> db.hobbits_SectionD.find();
[
  {
    _id: 'm',
    value: { totalWeight: 728, count: 9, avgWeight: 80.88888888888889 }
  },
  {
    _id: 'f',
    value: { totalWeight: 406, count: 6, avgWeight: 67.66666666666667 }
  }
]
test>
```

Question Dc)

Generating a gender wise average weight report using aggregation pipelines

Code:

```
//using aggregate pipelines
db.hobbits.aggregate(
    [
        {
            $group:
                { _id:"$gender",
                avgweight:{ $avg:"$weight"}
            }
        }
    ]
)
```

Output:

```
[
  { _id: 'f', avgweight: 67.66666666666667 },
  { _id: 'm', avgweight: 80.88888888888889 }
]
test>
```

Appendix A: Marking Sheet:

158.337 Project Marking Sheet (Part B – 17.5% course mark)
(Attach this page to your project report BEFORE you turn it in.)

(Please make sure you provide all the necessary details)

Oracle Account: Group 35

Group Member 1 - ID number, Name

23021700, Mohit Negi

Group Member 2 - ID number, Name

23013649, Zaahid Benjamin

Group Member 3 - ID number, Name

20001249, John Lavack

Group Member 4 - ID number, Name

(Grader's section, please do not write below this)

PART B: _____**/60marks**

Part B Comments: